

Econ 139: Labor Economics

Labor Supply (Theory)

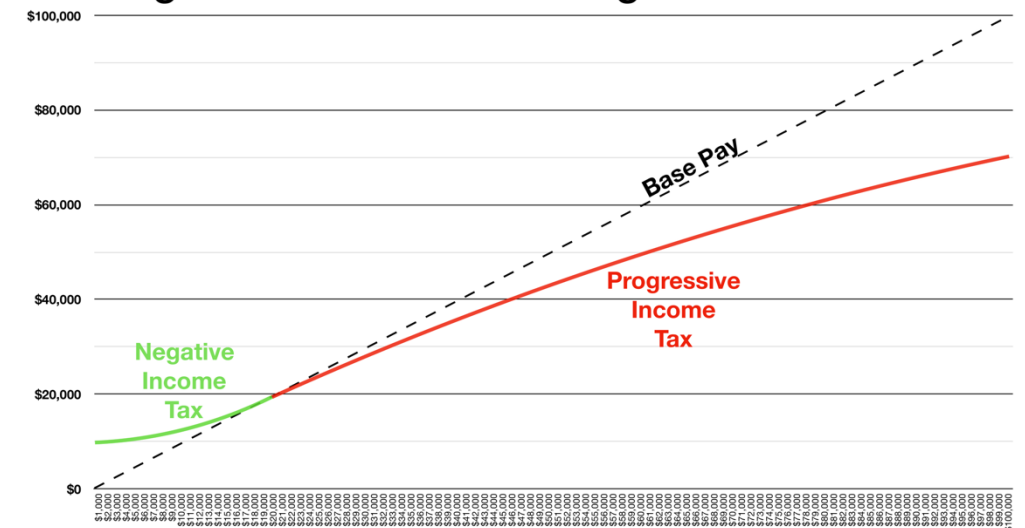
David Arnold

The Phase-In and Phaseout of the EITC

Credit Amount by Marital Status and Number of Children



Negative Income Tax & Progressive Income Tax



Labor Supply

- So far in the class, we have focused mostly on topics related to the firm's labor demand
 - How does technology impact firm's decision to hire workers
 - How does minimum wage change firm's decision to hire workers
- This week we will turn to the supply side of the market: the workers

Theoretical Model

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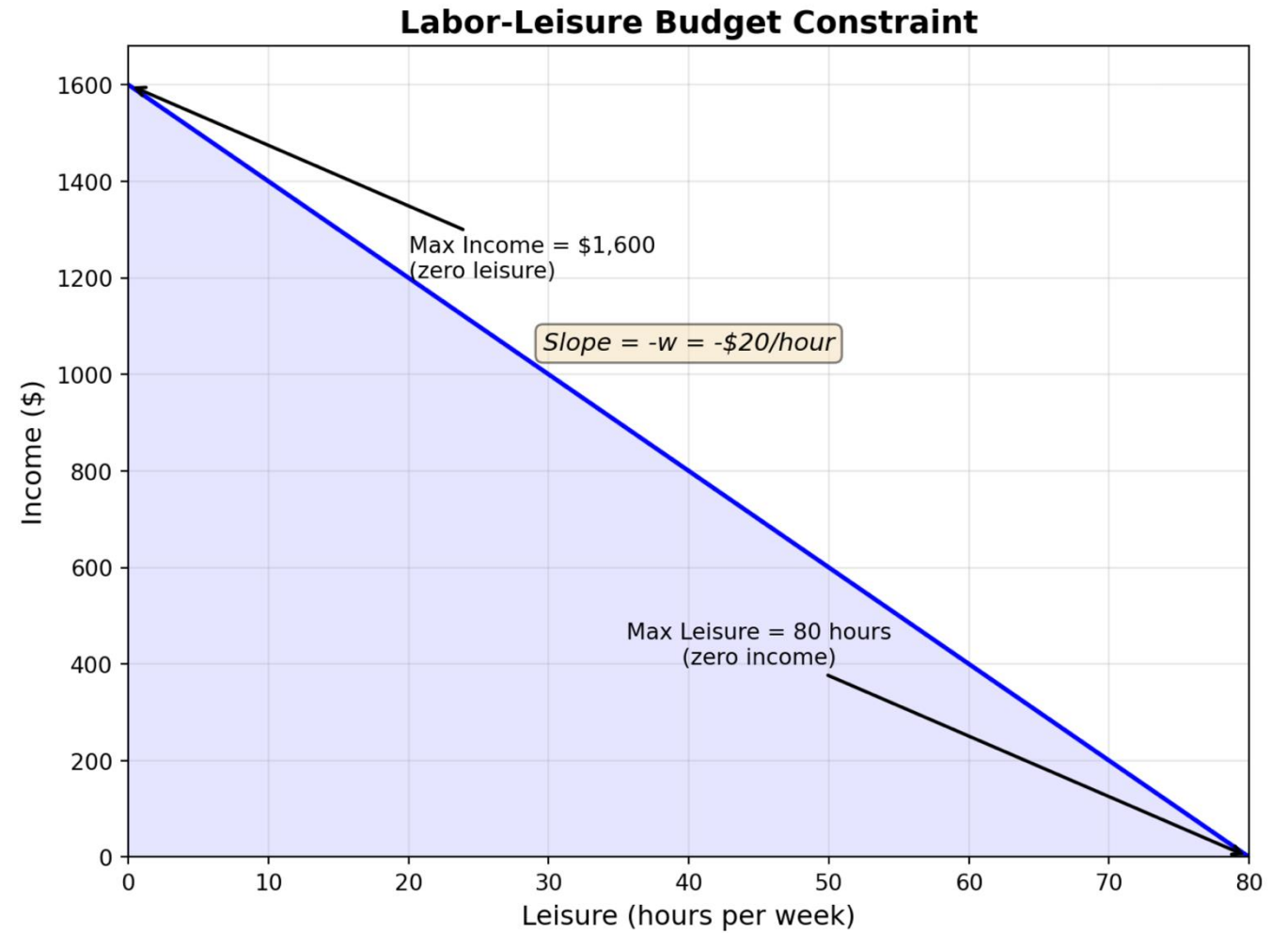
Theoretical Model

- The theoretical model we will study is relatively simple:
 - Workers face some wage rate w
 - They can choose how many hours to work (or equivalently, number of hours of leisure)
 - The tradeoff is between more income (which yields higher consumption) vs. more leisure time

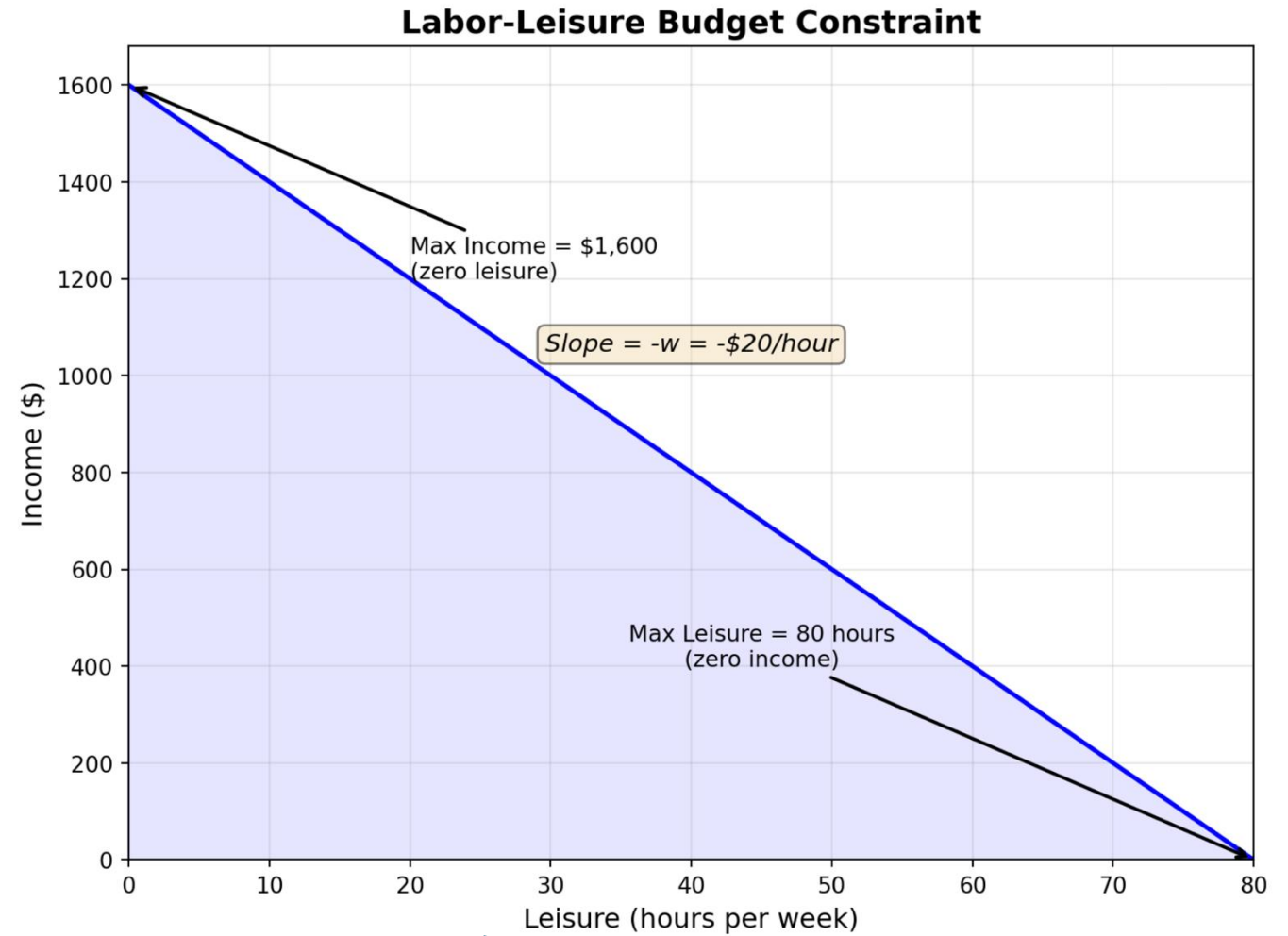
The Budget Constraint

- The budget constraint tells us what levels of income are possible for a worker facing a wage rate of w
- To be concrete, let's assume that the worker can decide between working 0 to 80 hours and face a wage rate of 20 dollars per hour
- Let's draw the budget constraint for this worker

The Budget Constraint



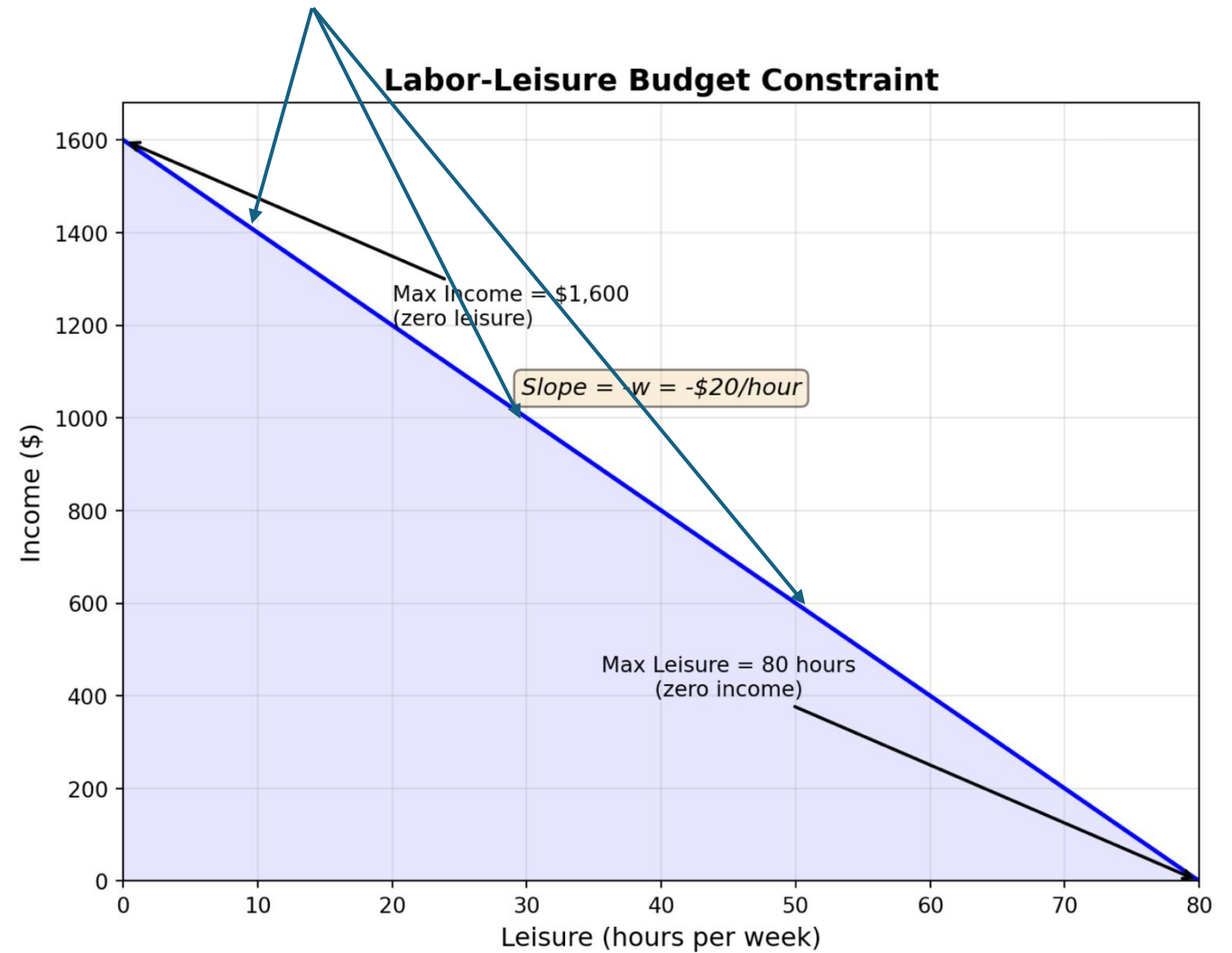
The Budget Constraint



Standard practice is to put leisure on the X-axis. Number of hours worked is equal to 80-leisure

How do we figure out which point a worker should pick? There are infinite choices (theoretically)

The Budget Constraint



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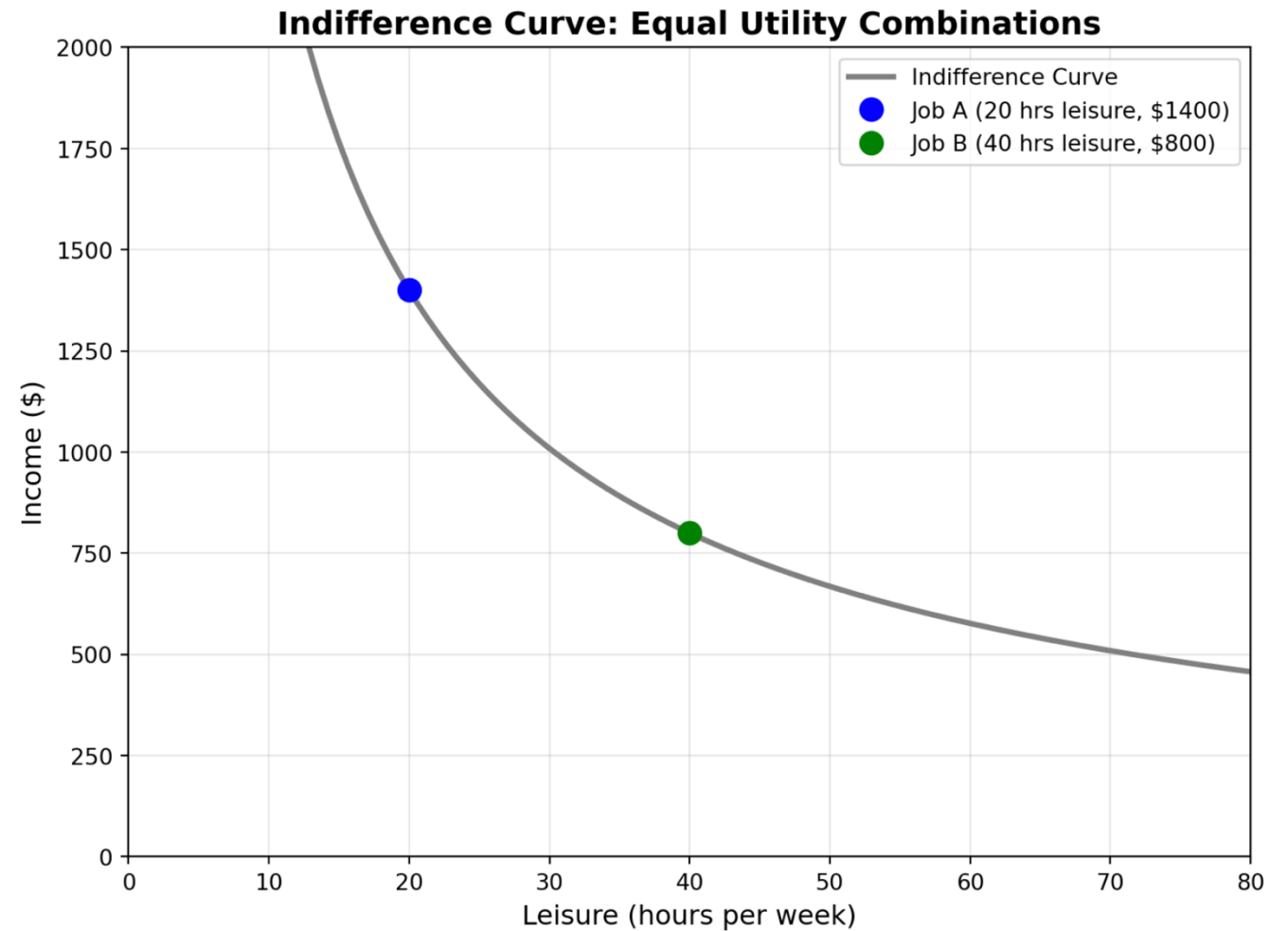
The Utility Function

- The optimal choice is the one that maximizes a worker's utility function
- The utility function is a mathematical abstraction. For each combination of leisure and income, we will think about that combination as producing a number that represents how much the individual likes that combination
- They will choose the combination with the highest utility

The Utility Function Example

- Imagine I have two job options
 - Job A: 60 hours and you receive 1400 in income
 - Job B: 40 hours and you receive 800 in income
- The job that you prefer yields higher utility for the worker
- However, if you were indifferent between the two jobs, we would say the two jobs lie on the same **indifference curve**

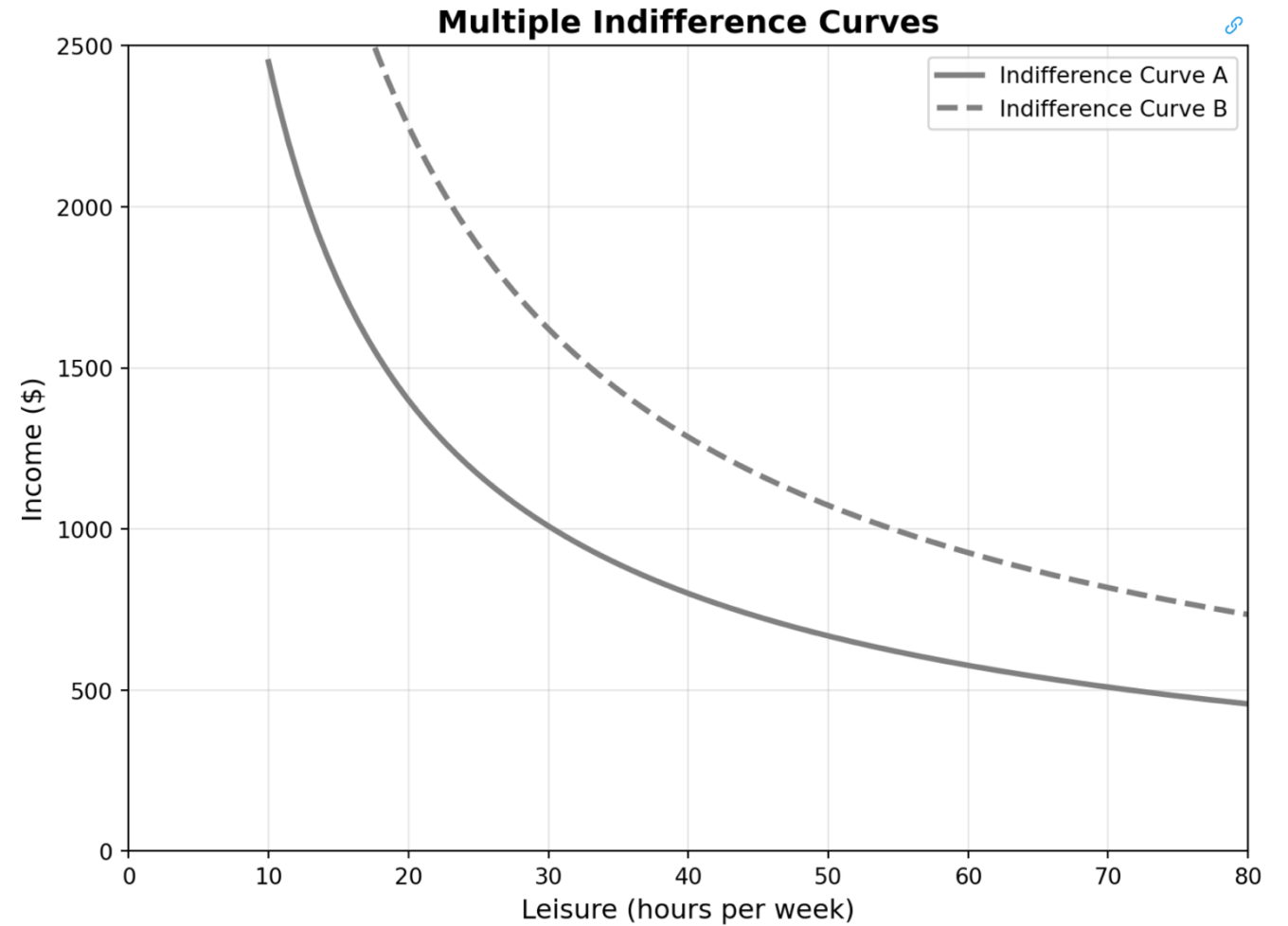
Indifference Curves



Any point on this curve yields the exact same utility, so the person is indifferent between these options

Indifference Curves

- Each person has (conceptually) infinite number of indifference curves
- In this example, curve A gives lower utility than curve B



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Indifference Curves Never Cross

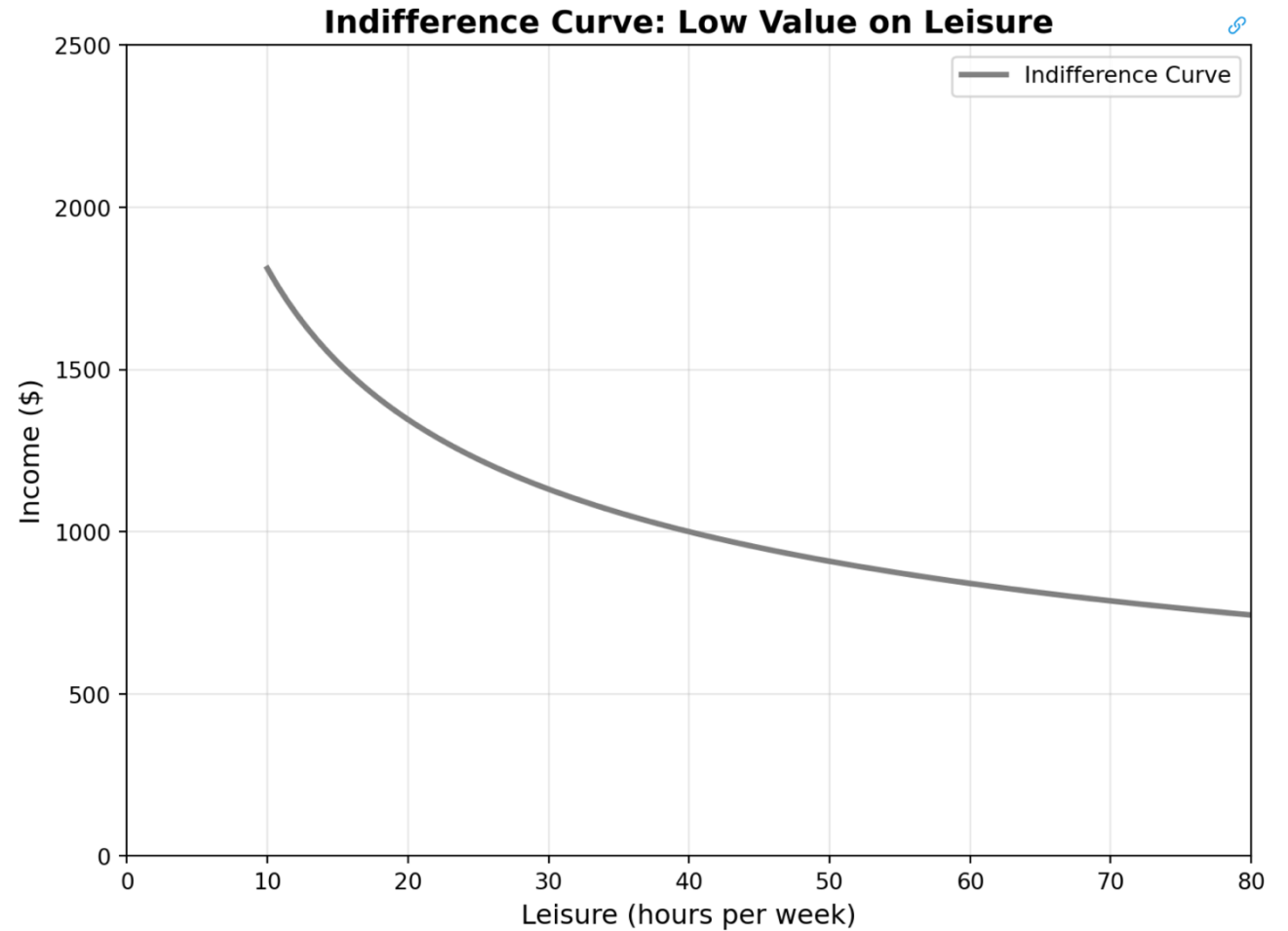
- Within an individual indifference curves can't cross.
- For the sake of argument, let's assume that there is an indifference curve C that yields utility of 100 and indifference curve D that yields utility of 200
- If they crossed, then there would be at least one combination of leisure and income that yields utility of both 100 and 200
 - But this doesn't make sense!
 - It would be like asking someone to rate a movie out of 5 stars and they tell you it is both 3 stars and 5 stars.

Preferences Determine Shape of Indifference Curve for a Person

- Different individuals can certainly have differently shaped indifference curves
- What determines this is how they value leisure against income
- Let's take an example

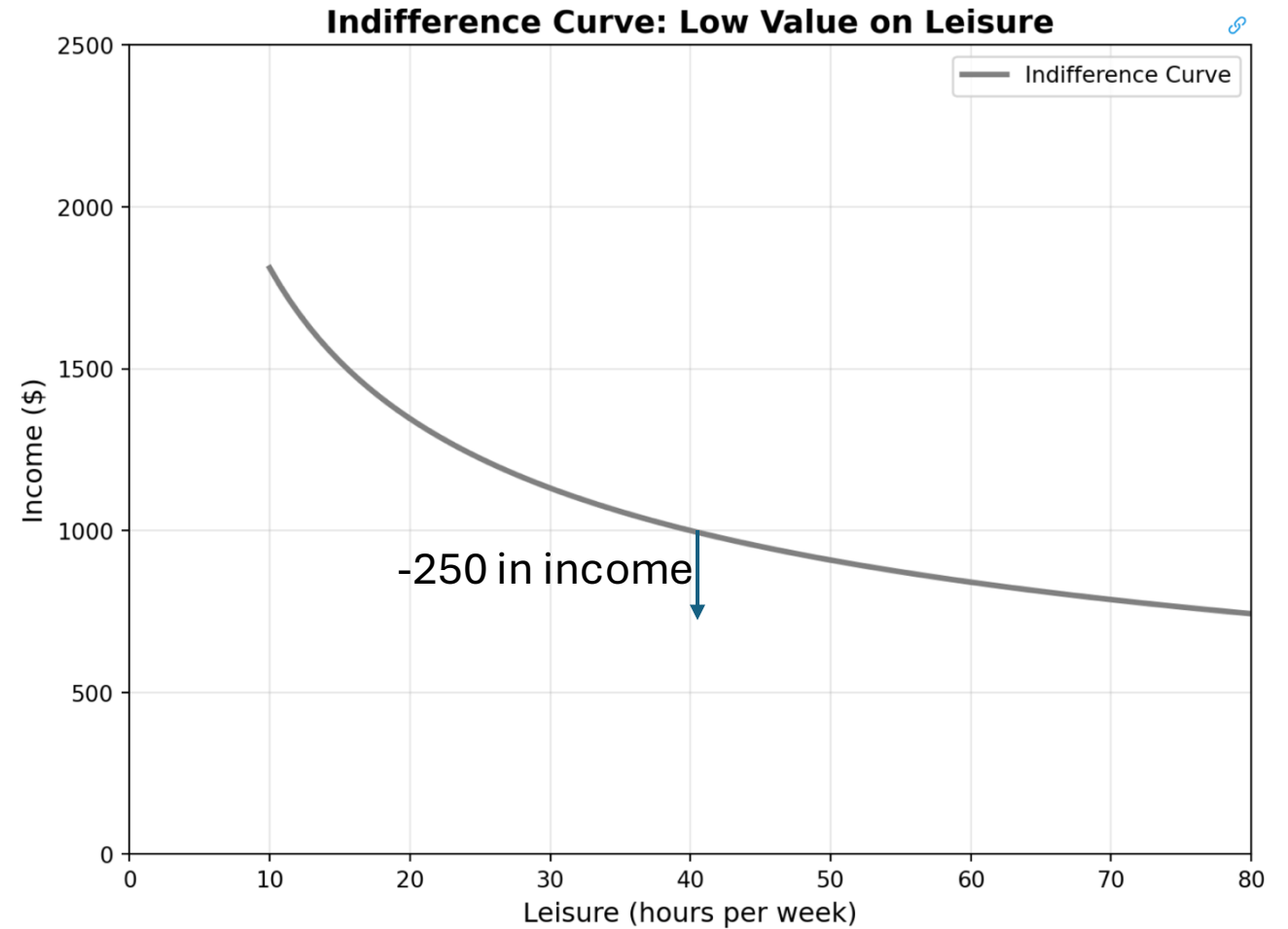
Preferences Determine Shape of Indifference Curve for a Person

- This indifference curve is relatively flat
- This implies that to be willing to give up a bit of income, you would need large change in leisure



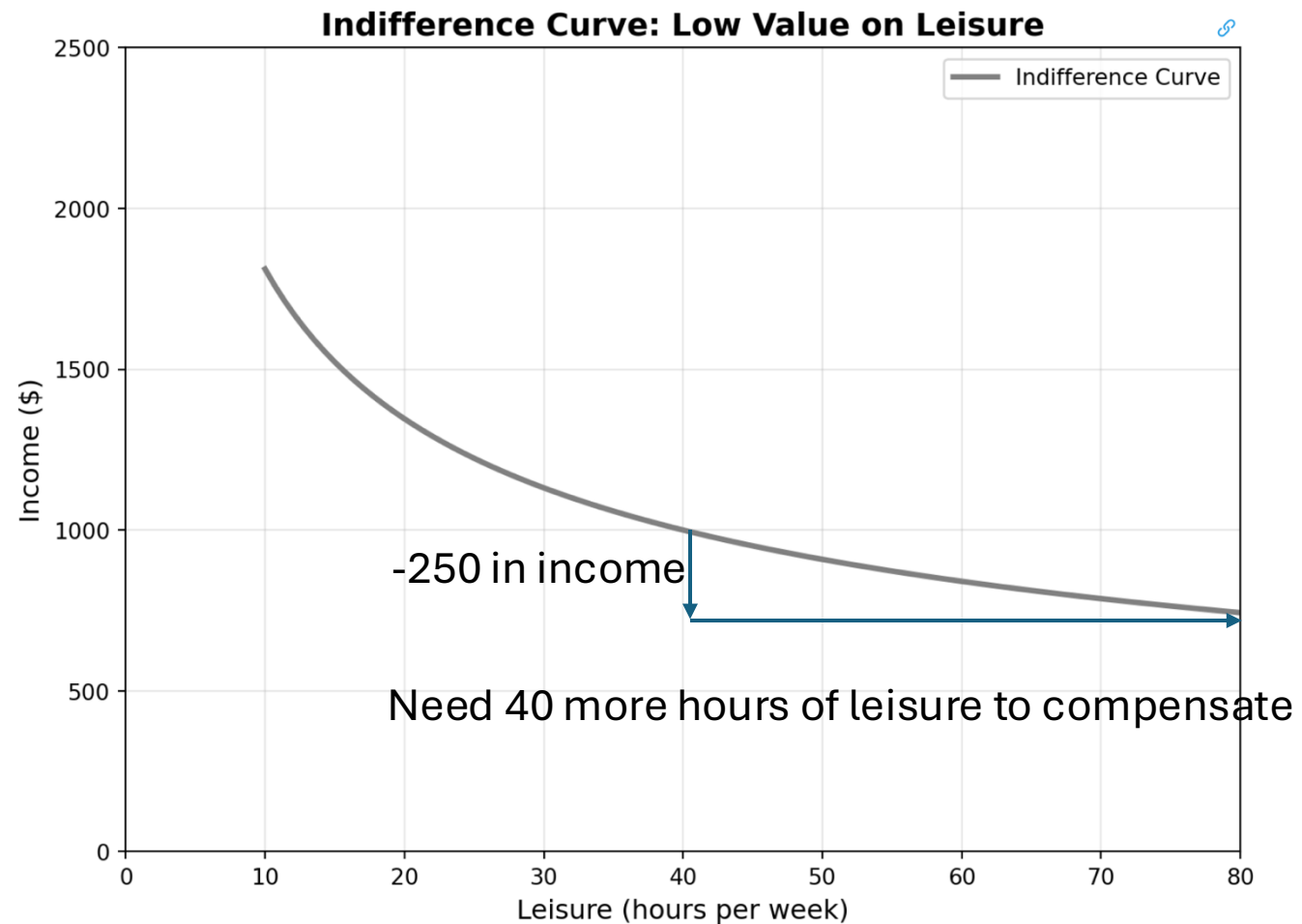
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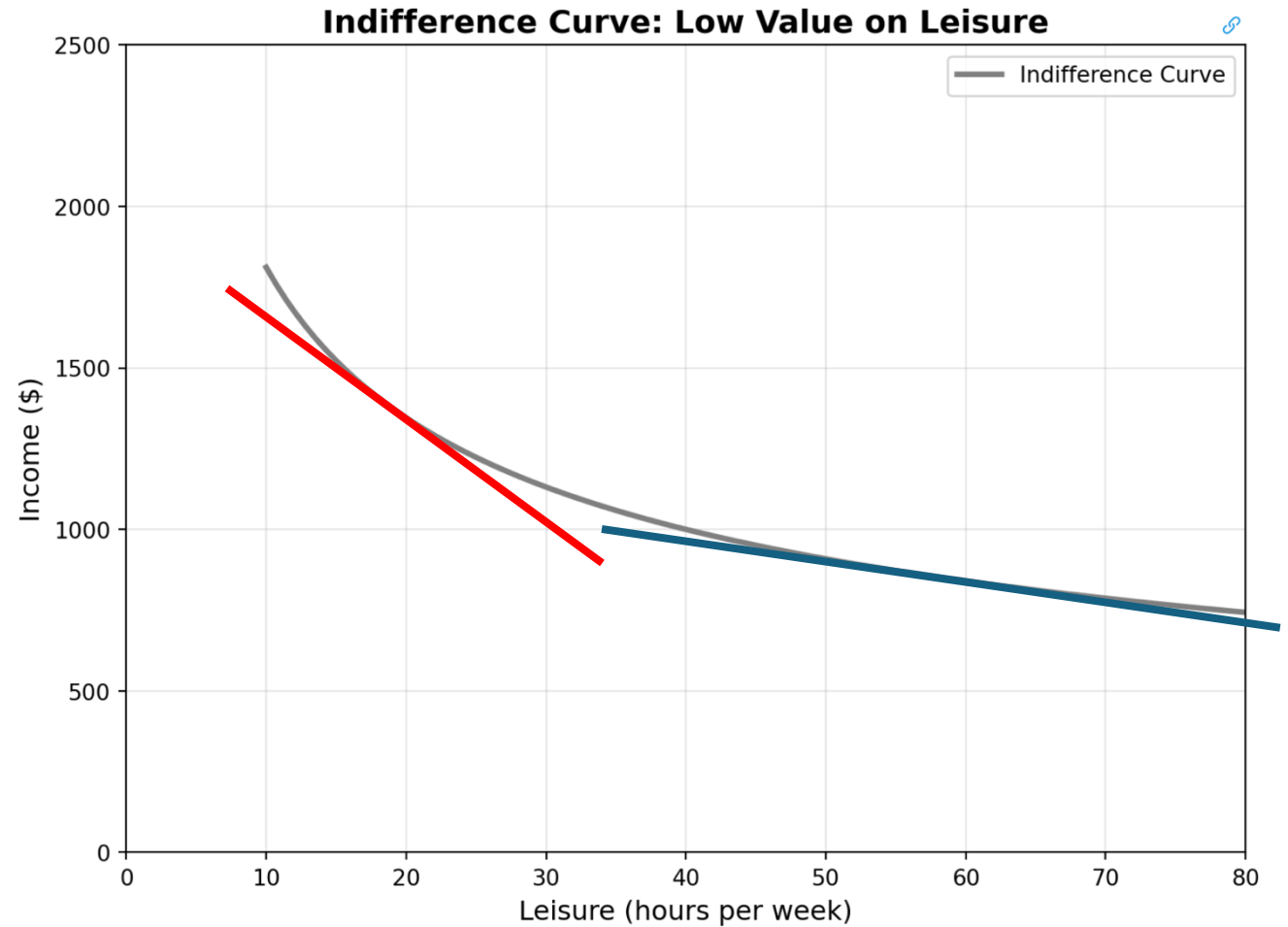
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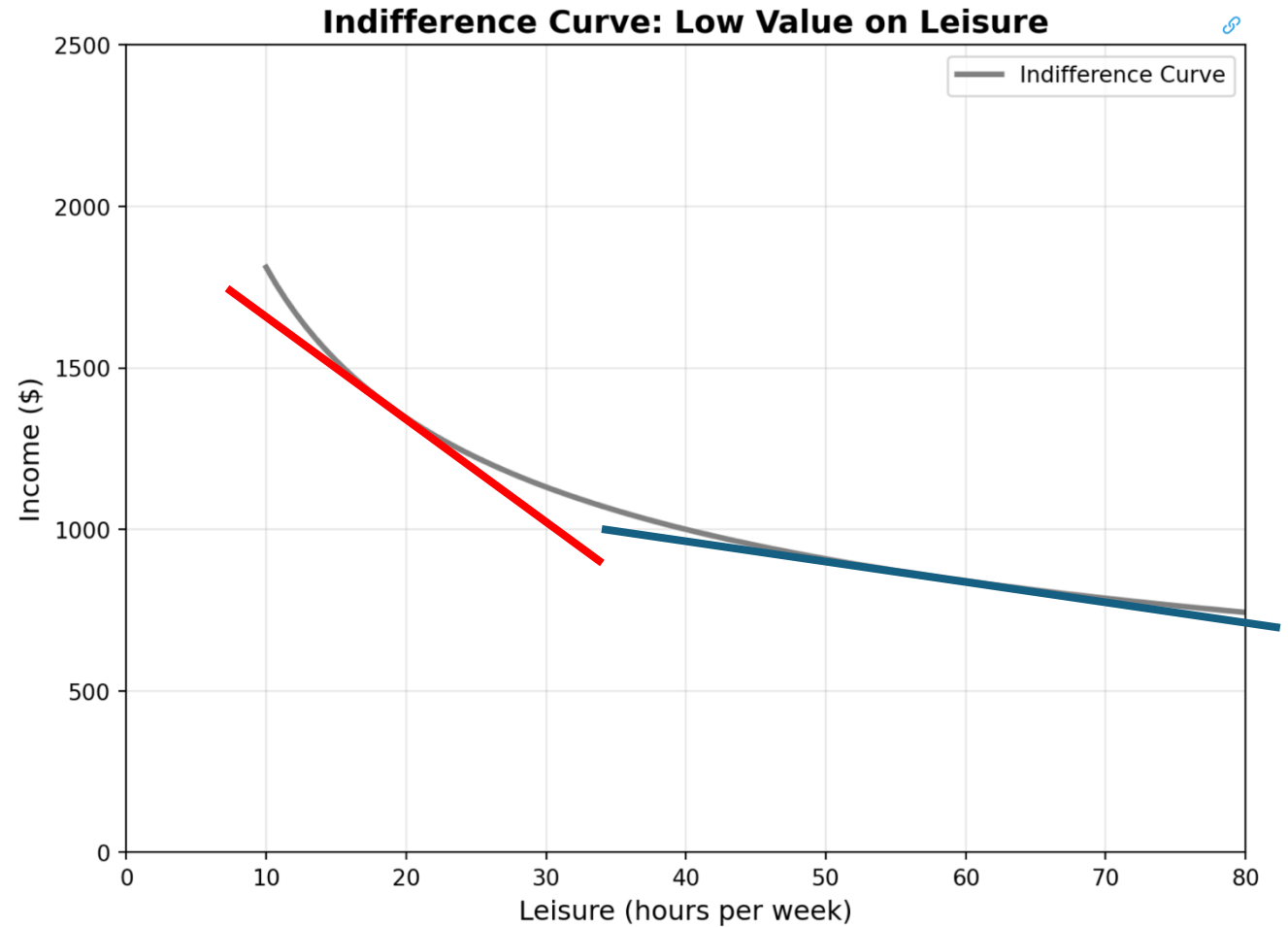
Slope of the Indifference curve

- The slope of the indifference curve is different at every point
- Slope is steeper at low values of leisure for this person
 - Willing to give up more income for smaller increases in leisure



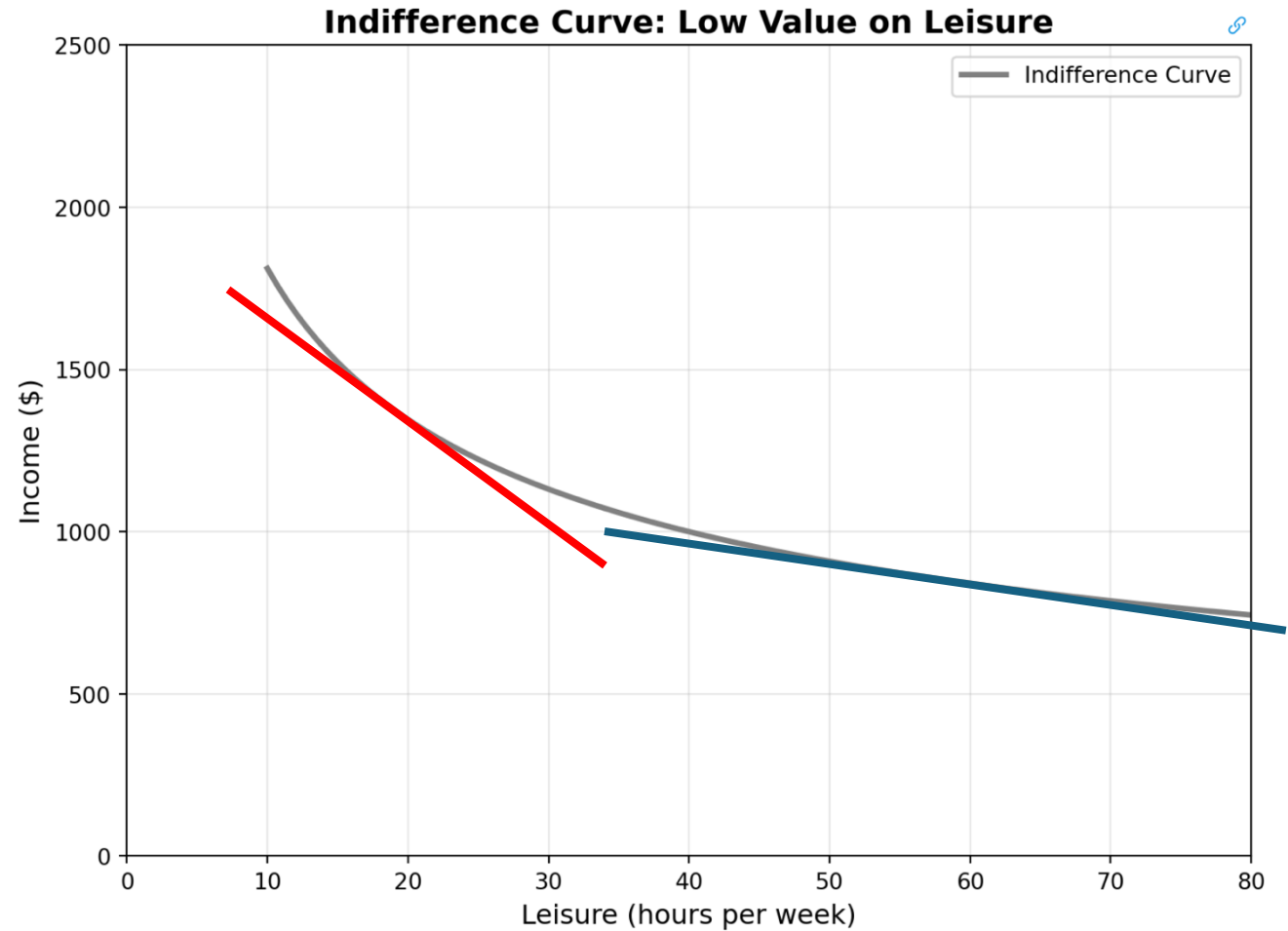
Slope of the Indifference curve

- The slope of the indifference curve is different at every point
- Slope is steeper at low values of leisure for this person
 - Willing to give up more income for smaller increases in leisure
- Slope is flatter at higher values of leisure
 - Willing to give up more income only for very large increases in leisure



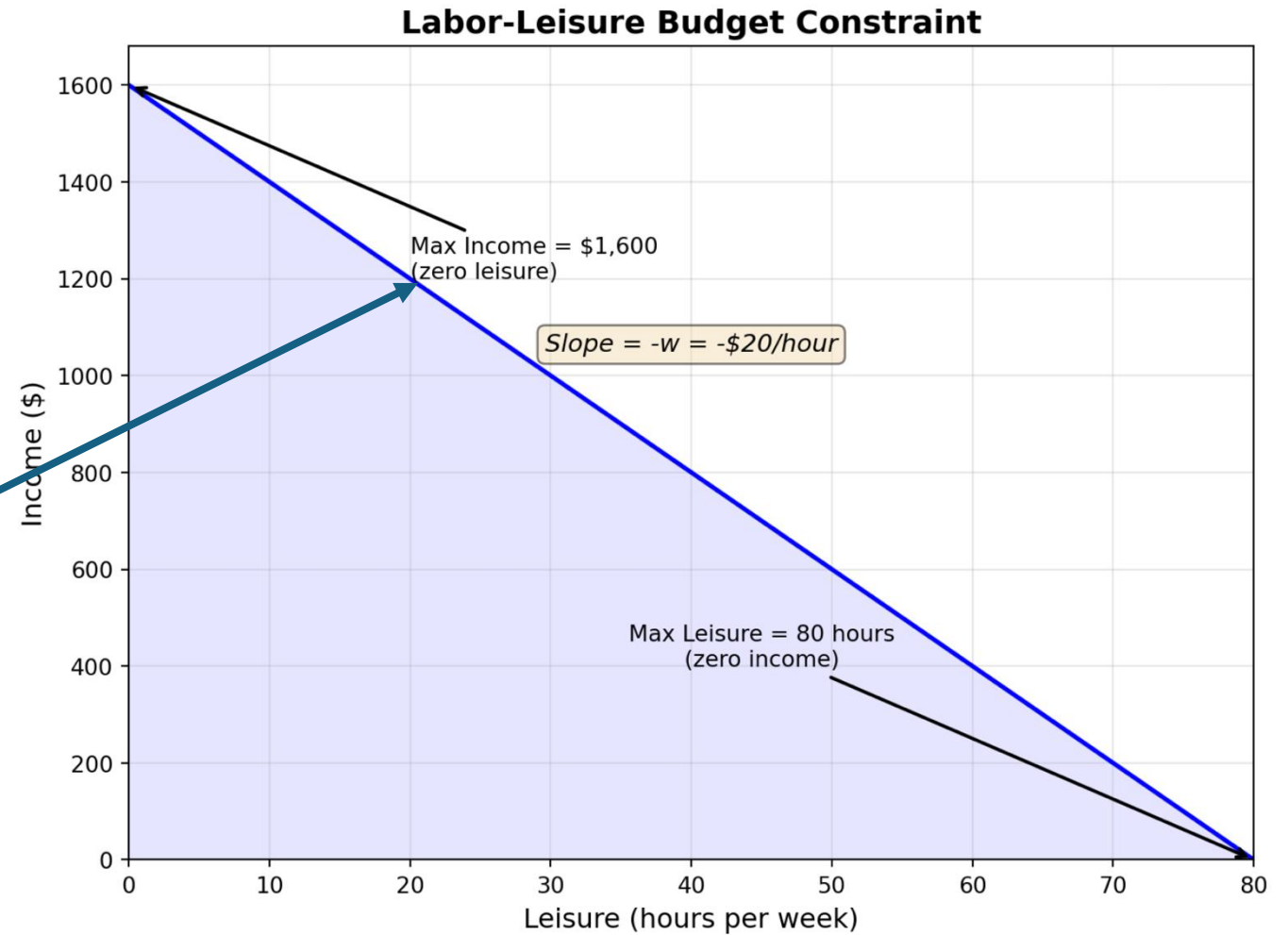
Marginal Rate of Substitution

- The slope at given point is referred to as the marginal rate of substitution
- It captures the tradeoff between income and leisure



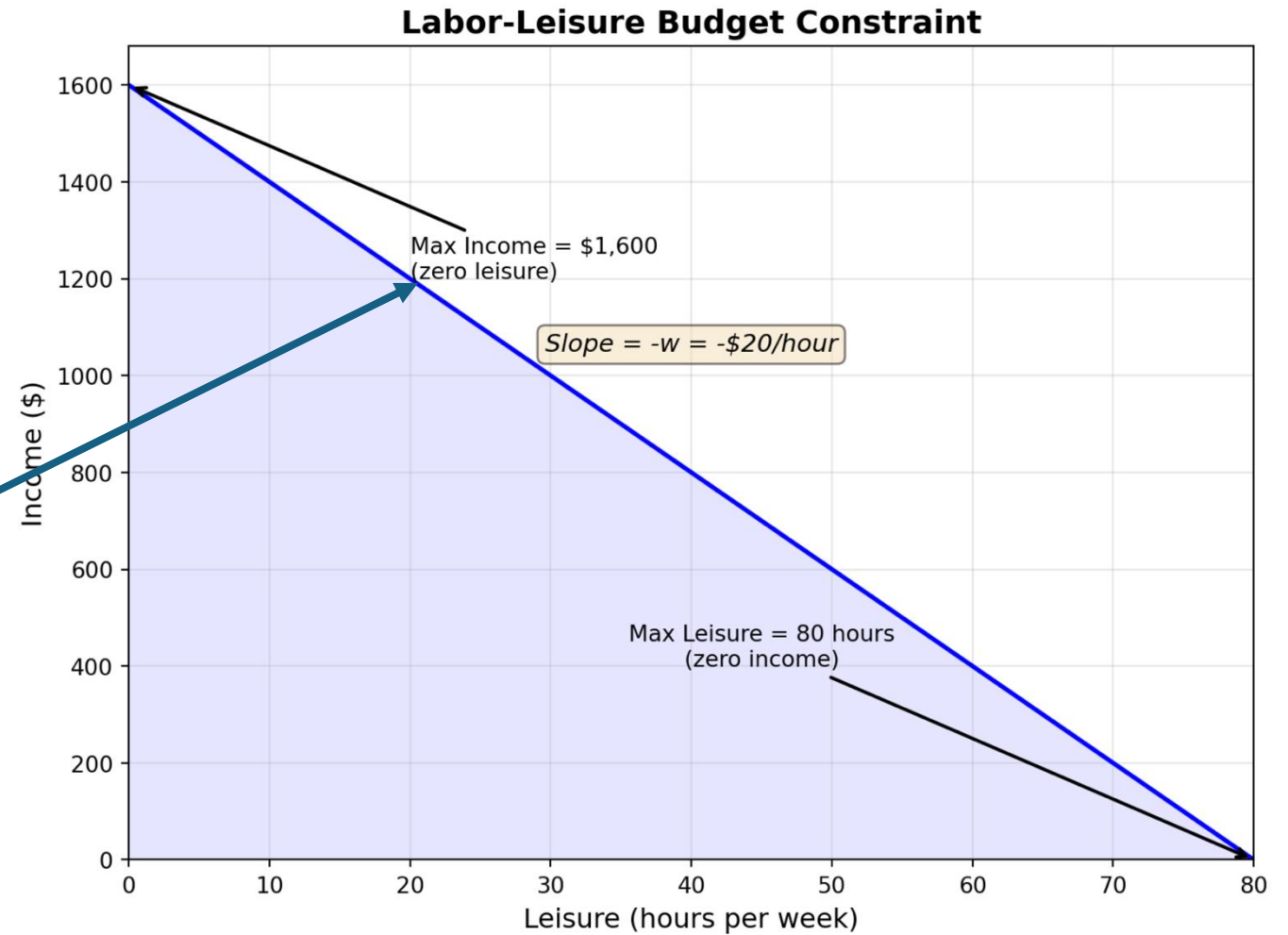
Optimal Level of Leisure

- To understand what the optimal choice for a worker looks like, let's just start with a choice that is possible given our original budget constraint (20, 1200)



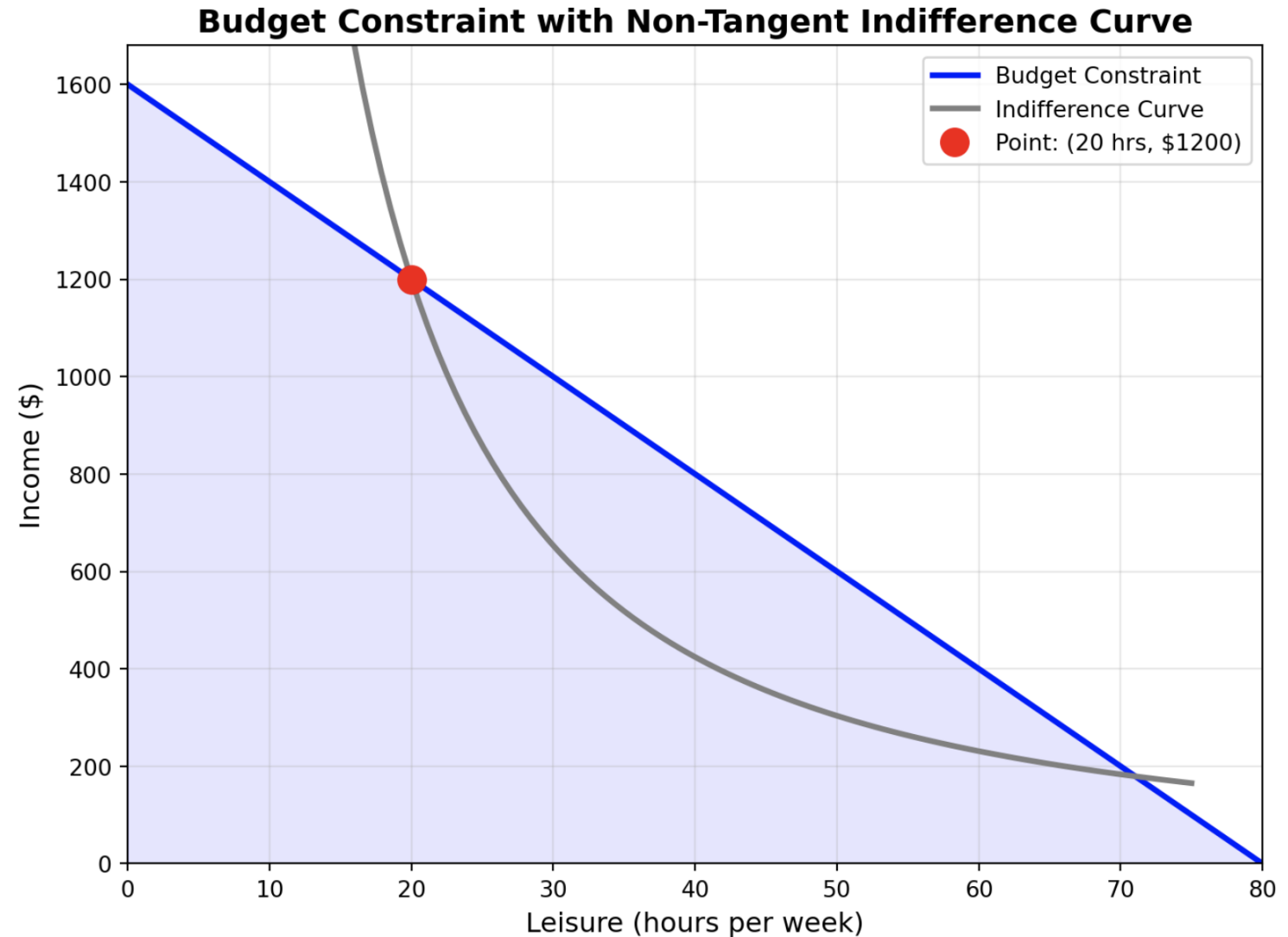
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- Let's now assume we know what the indifference curve that goes through this point looks like. Let's add it to the figure



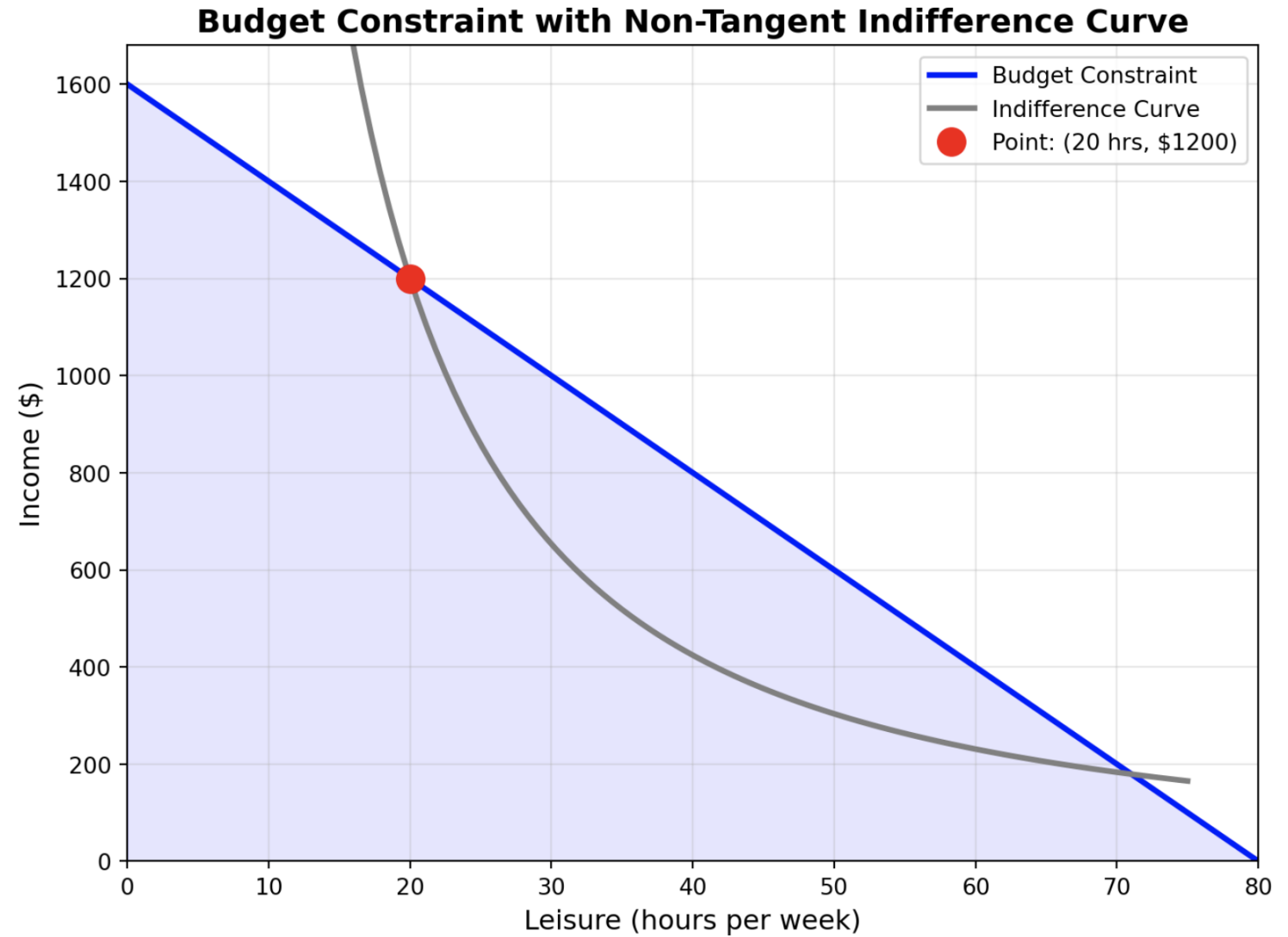
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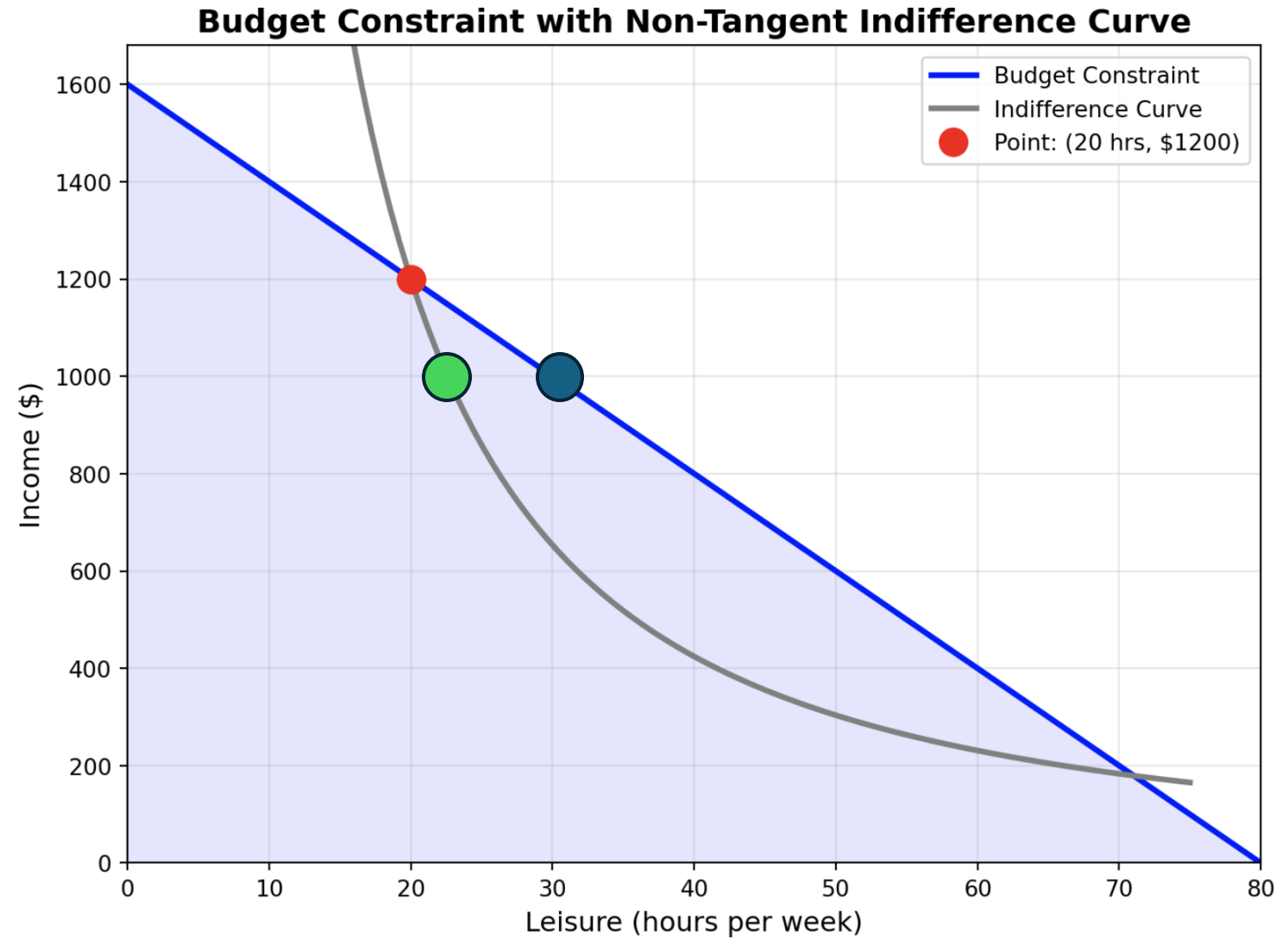
Optimal Level of Leisure

- Why is this not an optimal choice?



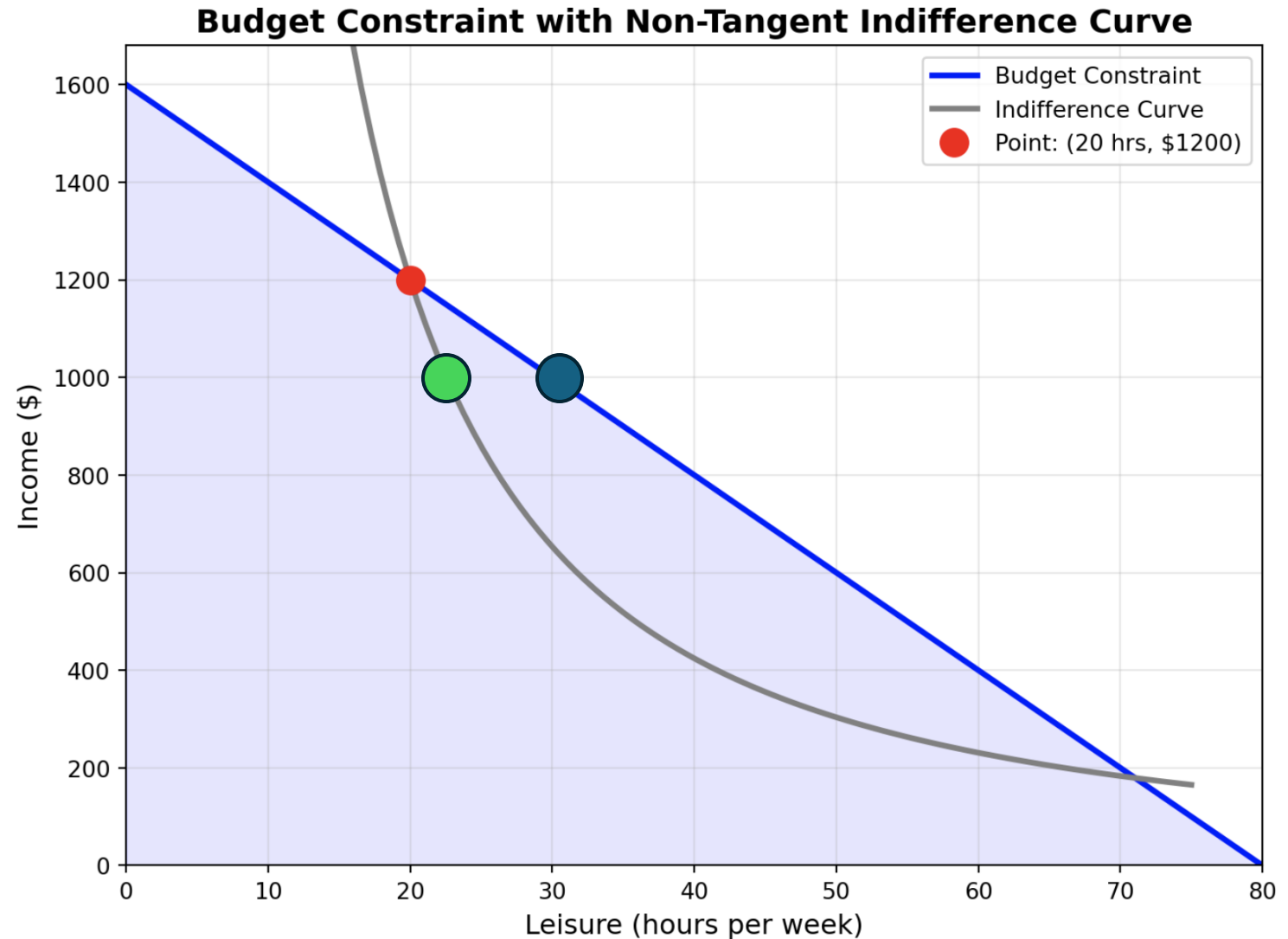
Optimal Level of Leisure

- Why is the **Red Dot** not an optimal choice
- (30,1000) [**Blue Dot**] is clearly preferred to (22, 1000) [**Green Dot**]. Same income, but more leisure with **Blue Dot**



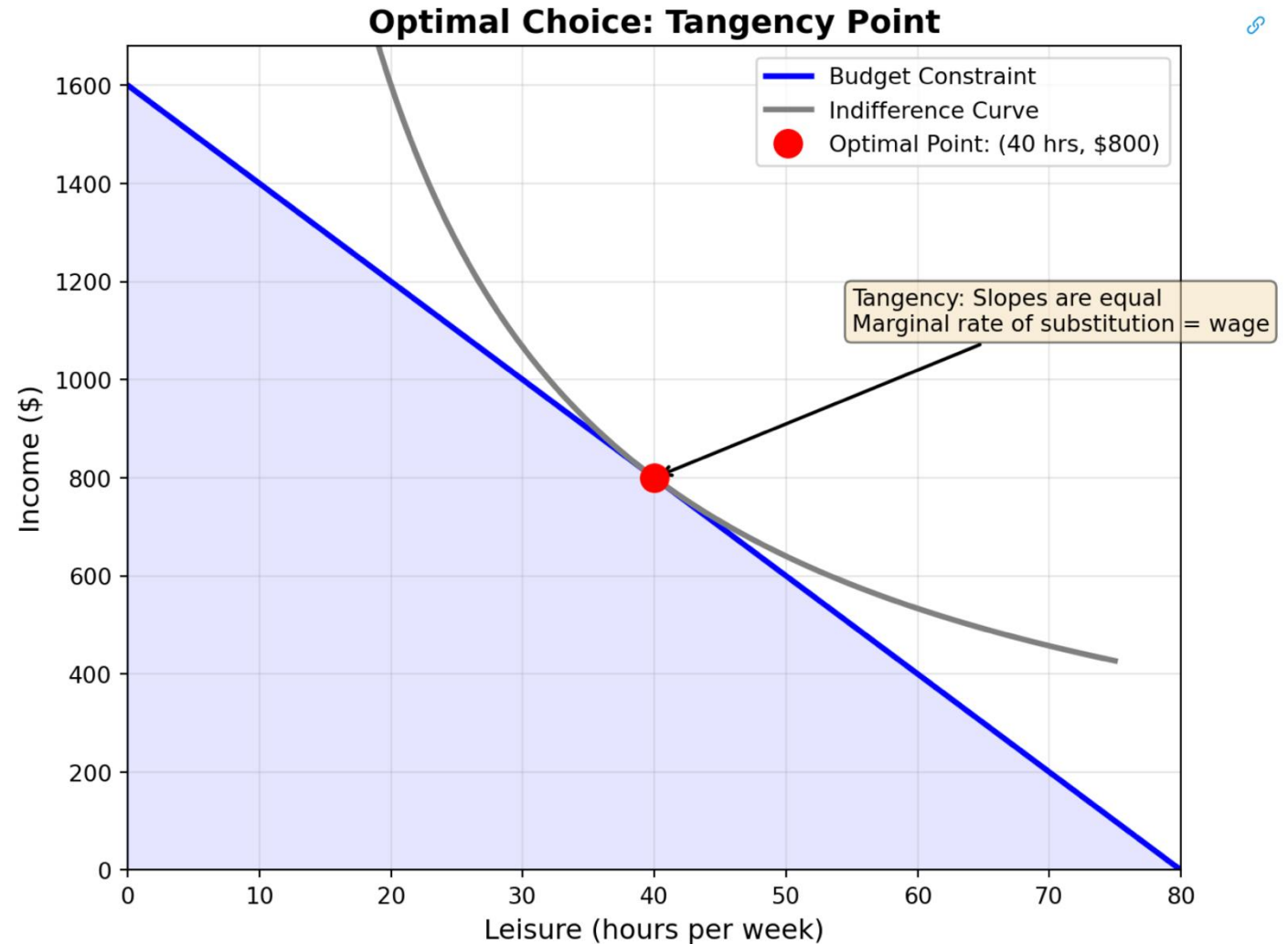
Optimal Level of Leisure

- Why is the **Red Dot** not an optimal choice
- (30,1000) [**Blue Dot**] is clearly preferred to (22, 1000) [**Green Dot**]. Same income, but more leisure with **Blue Dot**
- According to indifference curve, **Green Dot** is equivalent to **Red Dot**
- Therefore **Blue Dot** is preferred to **Red Dot**



Optimal Level of Leisure

- The optimal level of leisure is depicted now
- Intuition: If you want to move to higher indifference curve, you would now be off the budget constraint
- Mathematically: marginal rate of substitution between income and leisure is equal to the wage rate



Realism

- There are a few aspects of this simple model you may find unrealistic
- We assumed workers can choose **precisely** the amount of hours they would like to work
- However, in reality, there is often limited choice in hours worked and this is mainly set by firms
 - Worker preferences clearly do matter and can cause shifts in standard work hours (e.g. factory work in industrial revolution included more days per week and more hours per day than a standard 40-hour workweek in the US today)

Income vs. Substitution Effects

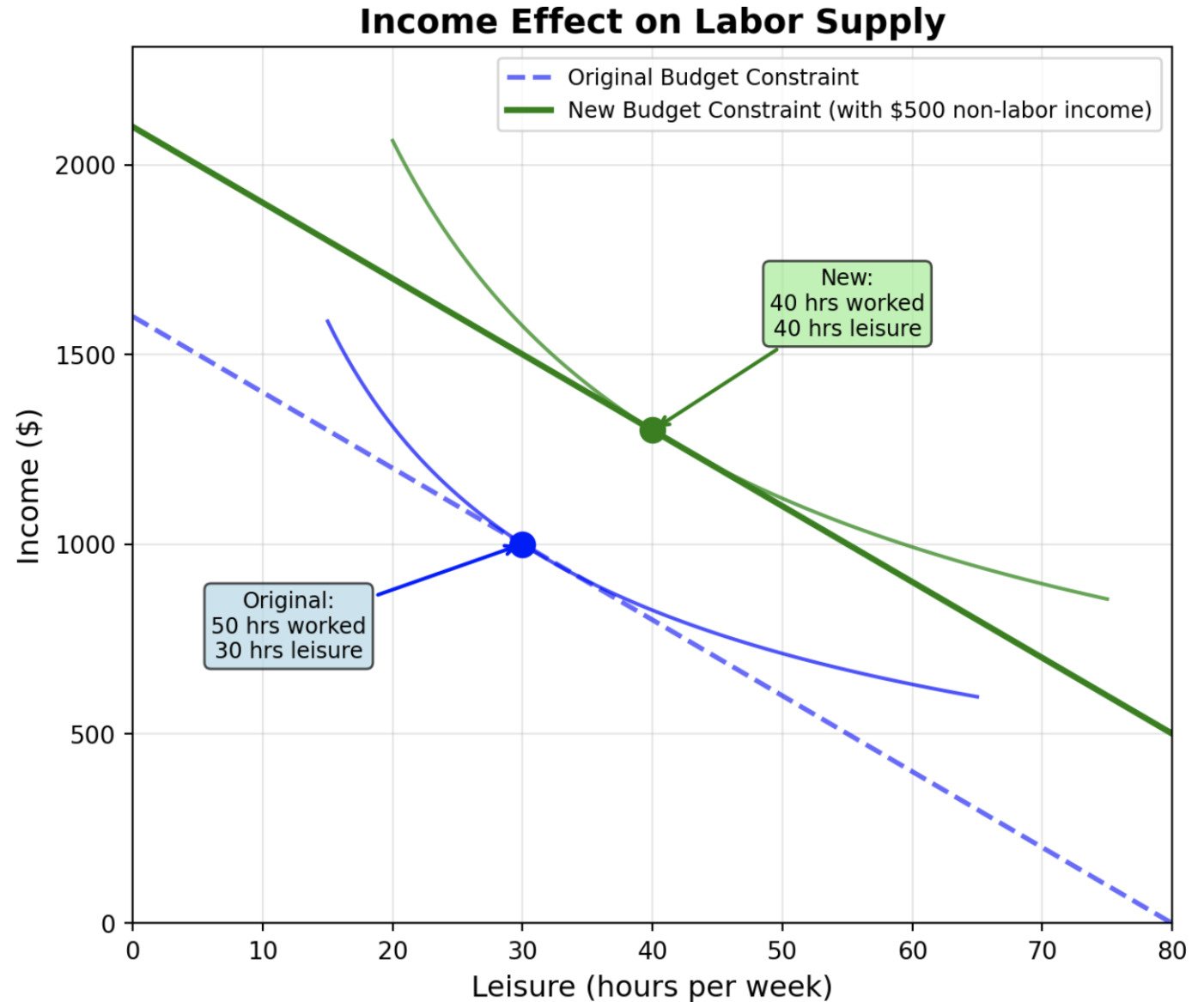
- A common distinction in labor supply analysis is the distinction between income and substitution effects, usually in the context of a wage increase
- When the wage goes up for an individual, there are two effects
 - **Income effect:** I now have higher income. This may make the worker want to work less
 - **Substitution effect:** I now receive more income per hour of work. This makes the worker want to work more
- Note that there is a somewhat unusual prediction here given our discussion of **market-level** labor supply curves: if the income effect dominates, you could get an increase in the wage resulting in lower demand for hours. While an interesting theoretical observation, I don't think there is good evidence for this occurring

Income Effects

- The exact definition of an income effect is: holding fixed the wage rate you face, how does your labor supply decision change in response to changes in income
- Your income could change (holding the wage rate fixed) for a variety of reasons
 - Rare (but useful for identification): lotteries
 - Common policy-relevant: changes in welfare benefits. These often have more complicated structures that may imply both income and substitution effects
- Let's see how this might be reflected in our graphs

Income Effects

- Theoretically, we believe for most individuals preferences, increases in non-labor income would result in fewer hours worked
- We can depict this situation with indifference curves
- However, theory doesn't tell us what this will look like on average (depends on people's preferences)



Income Effects

- Mathematically, we describe an income effect as a change in hours with respect to a change in income holding the wage rate constant

$$\text{Income Effect} = \frac{\Delta H}{\Delta I} | \text{Wage Constant}$$

Substitution Effects

- Substitution Effect is a change in hours with respect to a change in the wage rate holding the utility of a person constant

$$\text{Substitution Effect} = \frac{\Delta H}{\Delta w} | \text{Utility Constant}$$

Substitution Effects

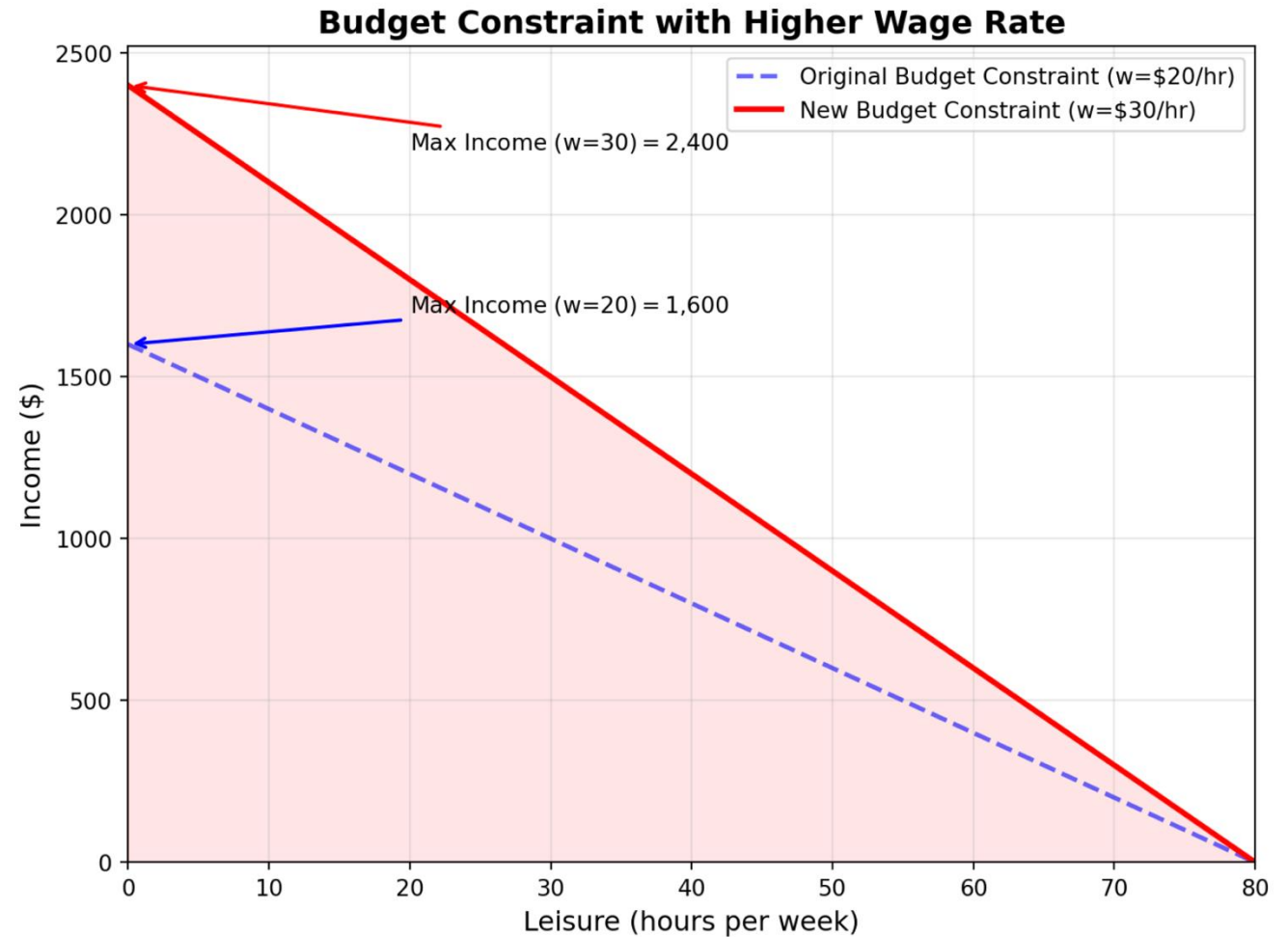
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- It is harder to think about a situation in which you are holding utility constant but changing the wage rate. Even conceptually, we never observe utility, so this would be hard to think about
- What we will show is that you can think about the change in labor supply due to a wage change as a combination of income and substitution effects

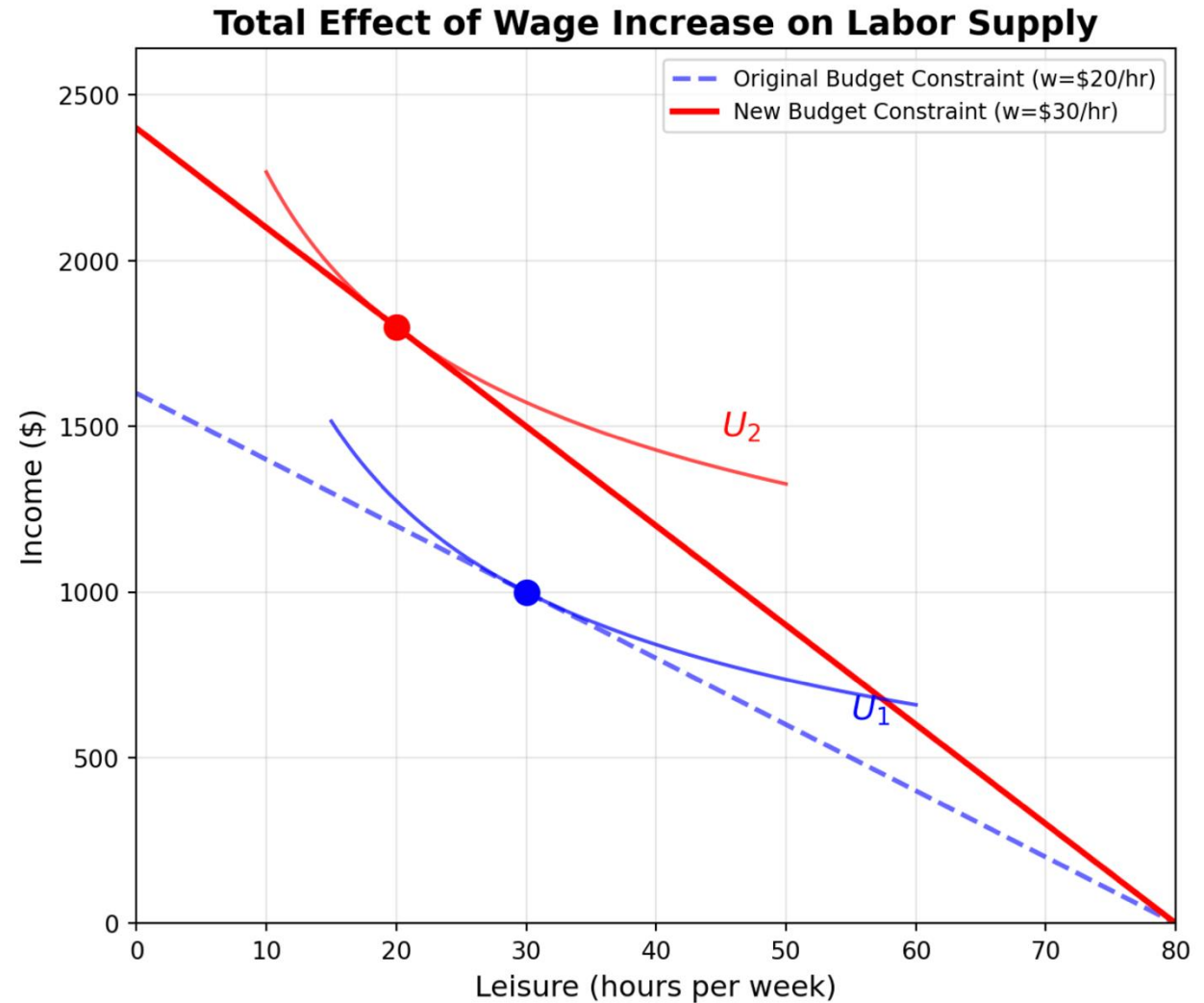
Decomposing Income and Substitution Effects

- Imagine wage rate increases from 20 to 30
- Let's assume the individual has preferences that make the following two decisions optimal



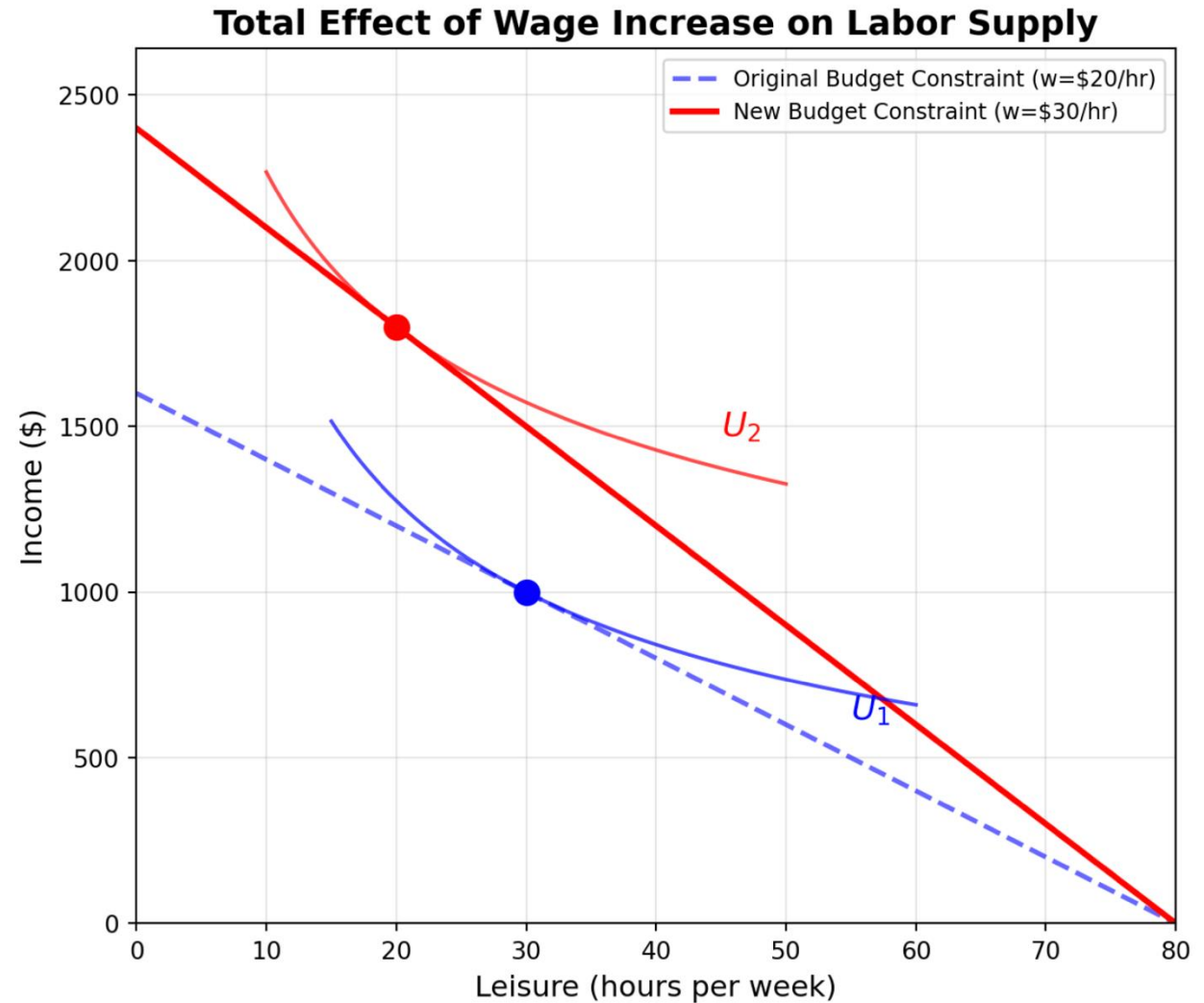
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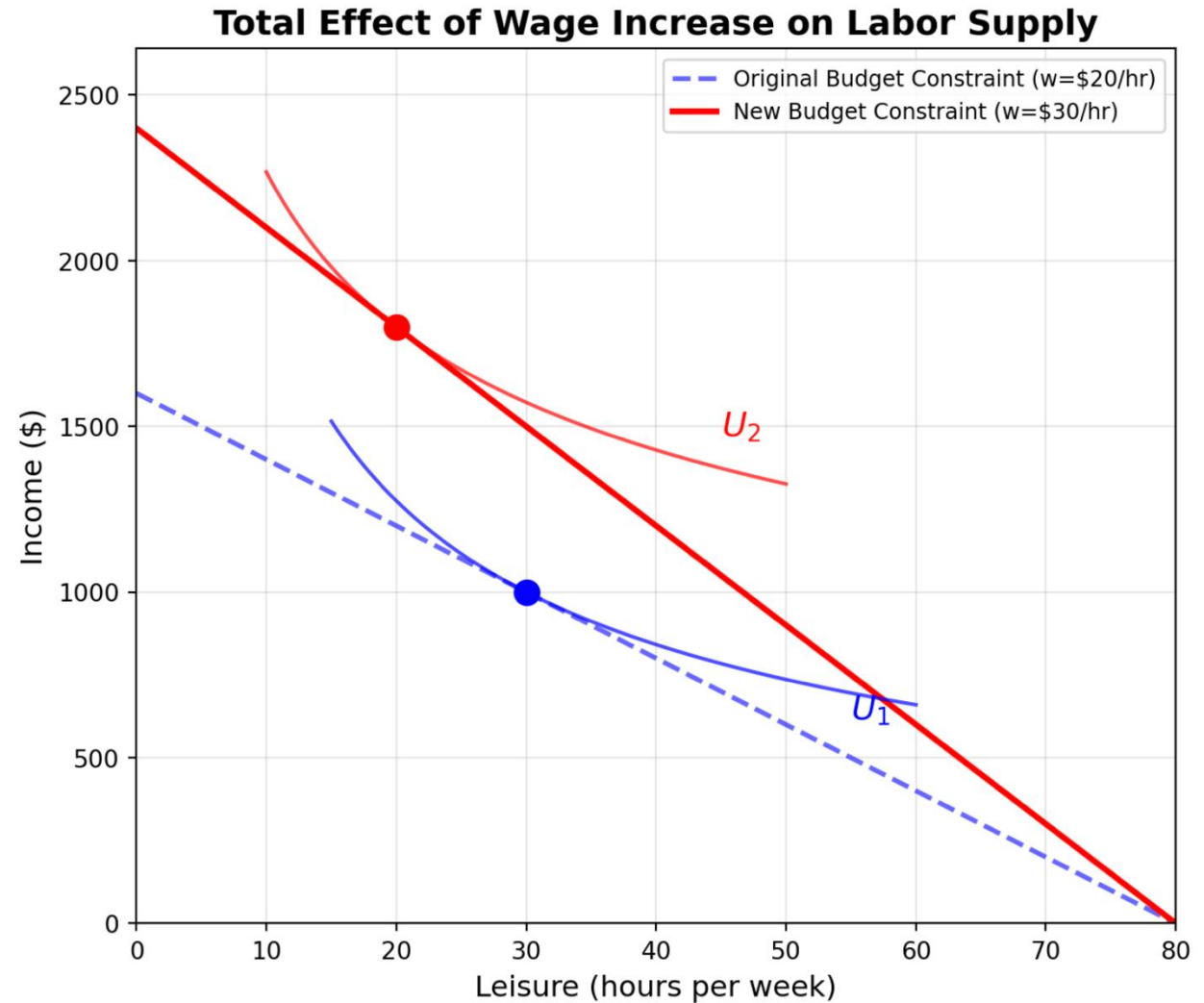
Decomposing Income and Substitution Effects

- Wage rate **decreased** leisure from 30 to 20 hours
- Part of this will be due to a
 - Substitution effect: Will always decrease leisure
 - Income effect: Will always increase leisure
- Let's isolate the income effect



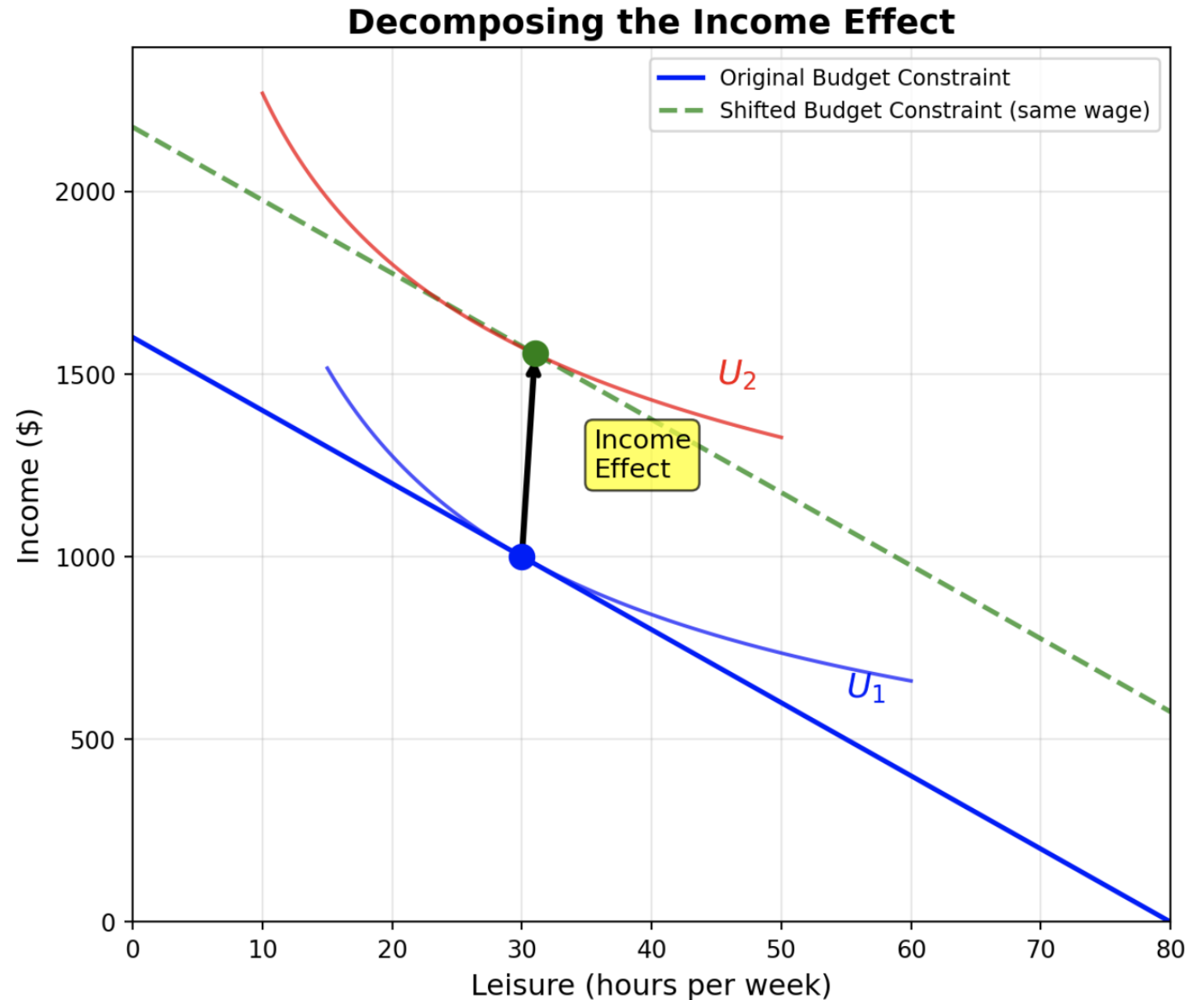
Isolating Income Effect

- Hypothetical question: if I want to get this worker from U_1 to U_2 , but with the same wage rate, what would be the optimal hours worked?
- The way I can do this is by shifting up the original budget constraint (i.e. increasing non-labor income) until it is tangent with U_2



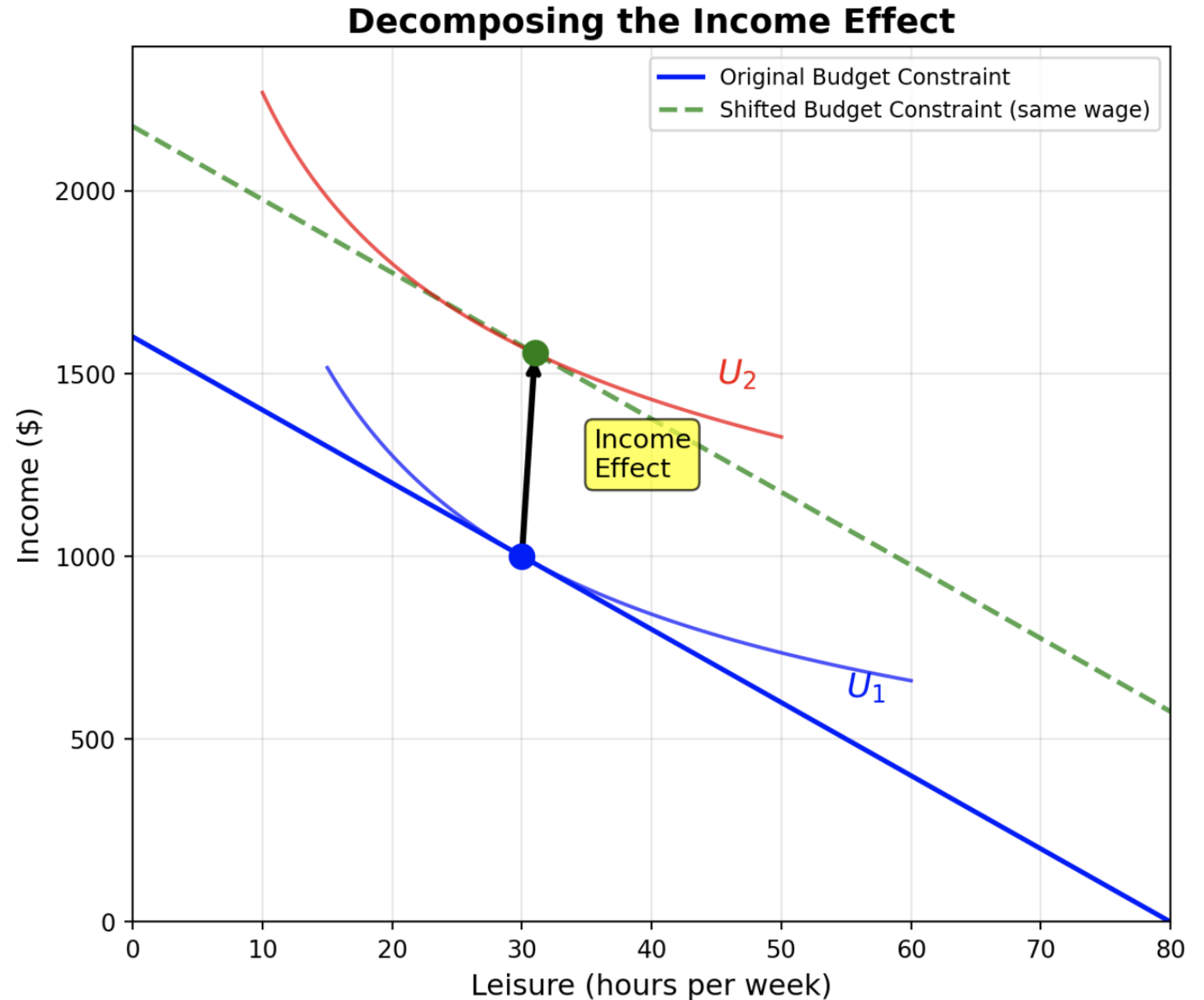
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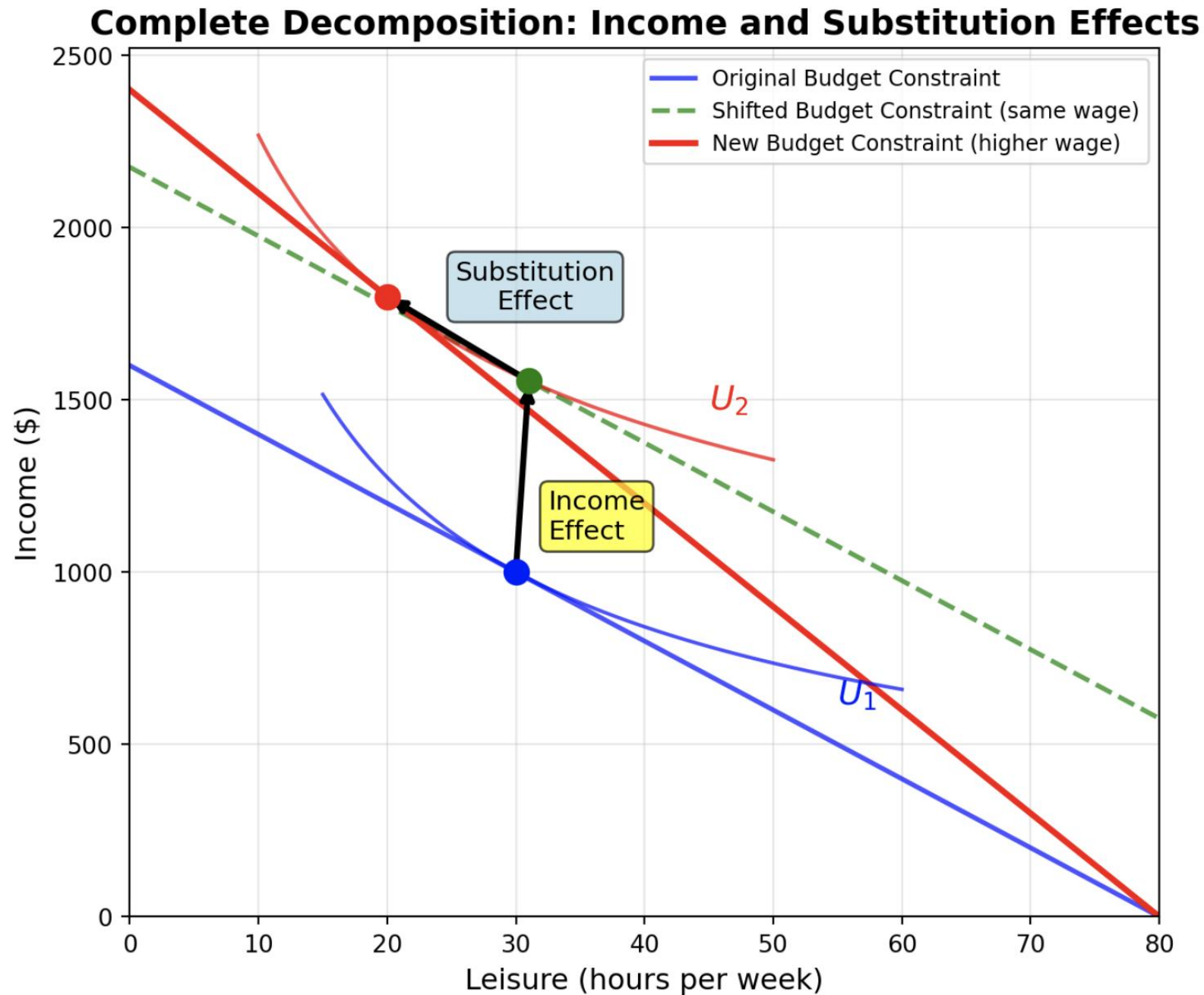
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- Now, substitution effect is change between green dot and the actual choice
- Why? Because this is the change in hours changing the wage rate, but **keeping utility fixed at U_2**



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Conclusion

- Theoretical model gives us a way to reason through how workers make labor-supply decisions
- Next class we will turn to empirics
 - Income effects have been getting some renewed interest as universal basic income proposals have seen a rise in popularity
 - We will revisit some older work on this which introduced some of the first field experiments in Economics, before turning to some more recent work and discussing long-run trends in labor supply